KEYENCE

Safety Interlock Switches

NEW GS-M/GS Series

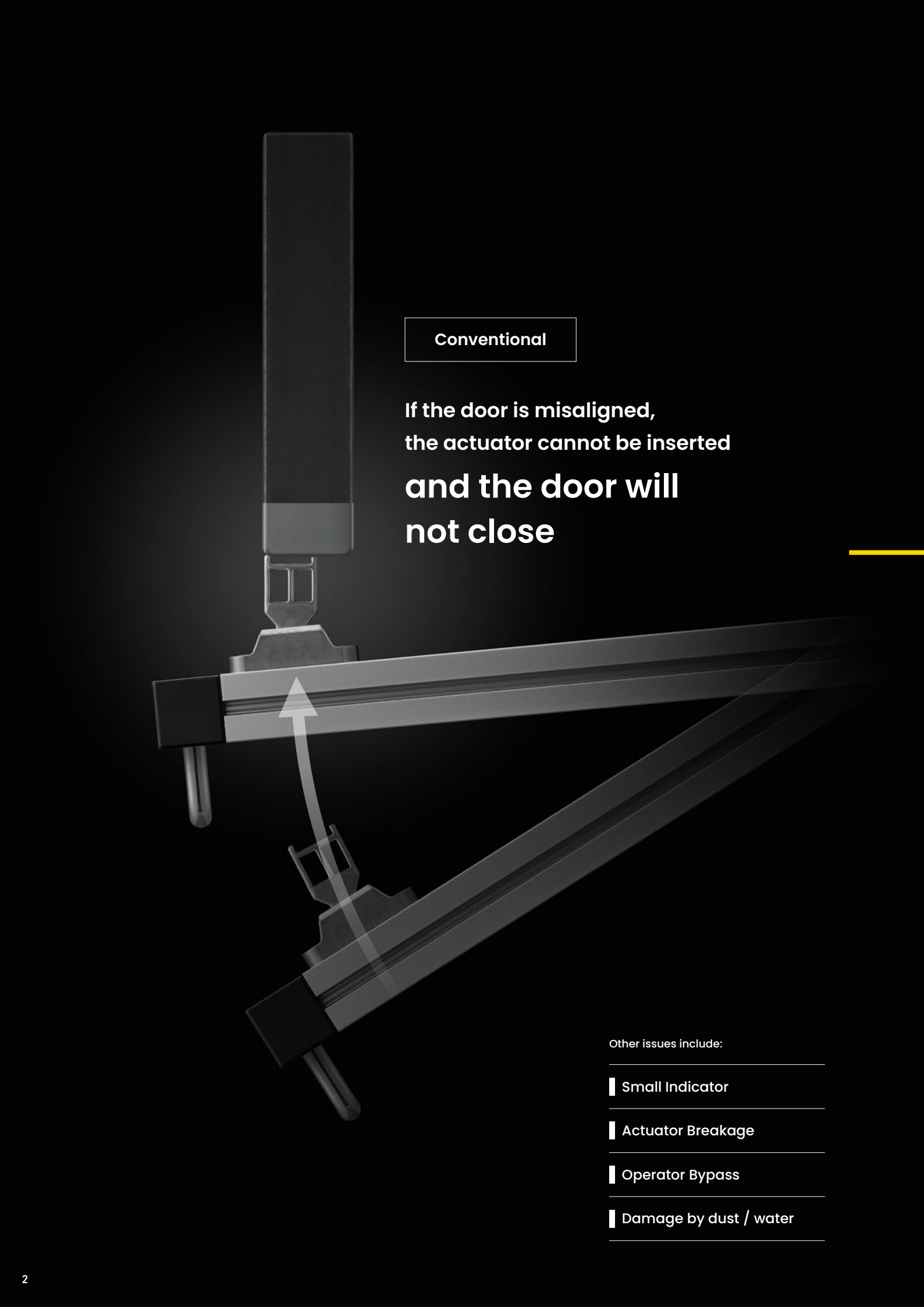


Machine Safety Made Simple

Achieve the Top Safety Standards

PLe | Category4 | SIL3

CE TÜV SÜD LISTED



Conventional

If the door is misaligned,
the actuator cannot be inserted

and the door will not close

Other issues include:

- Small Indicator
- Actuator Breakage
- Operator Bypass
- Damage by dust / water

GS-M Series

The door will lock without needing to insert the actuator,

solving the problem

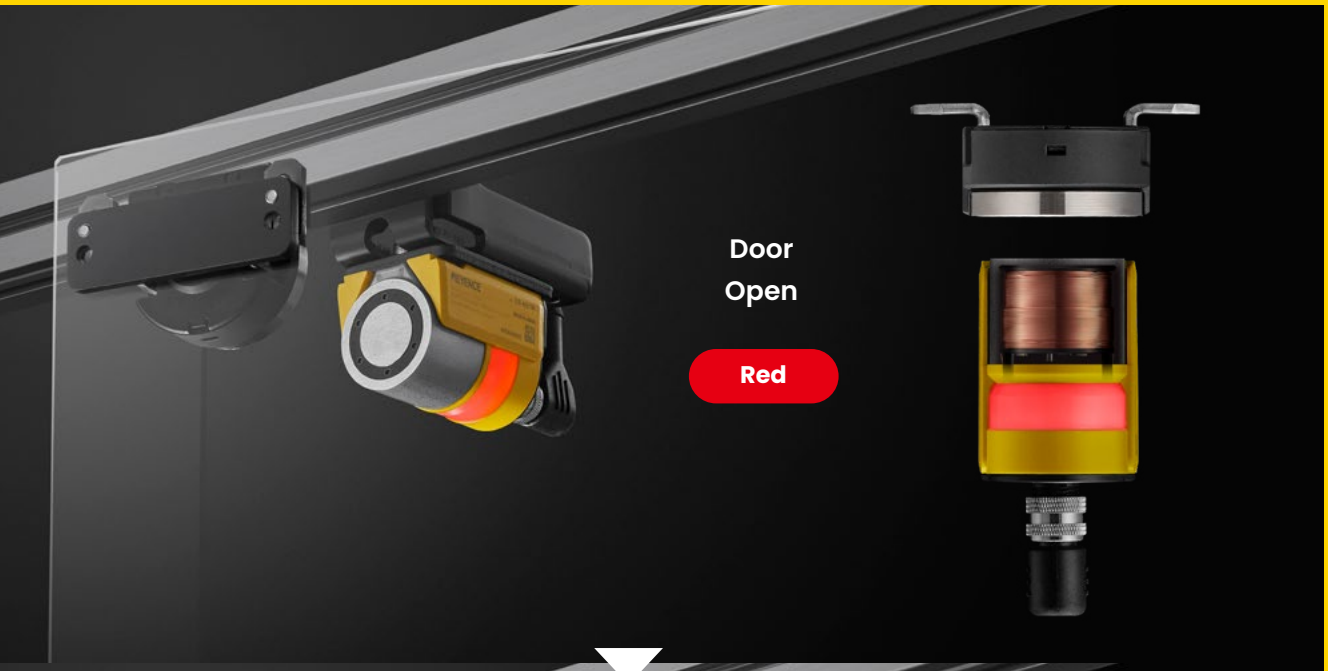


The latest with Super Small Body & Electromagnet Lock



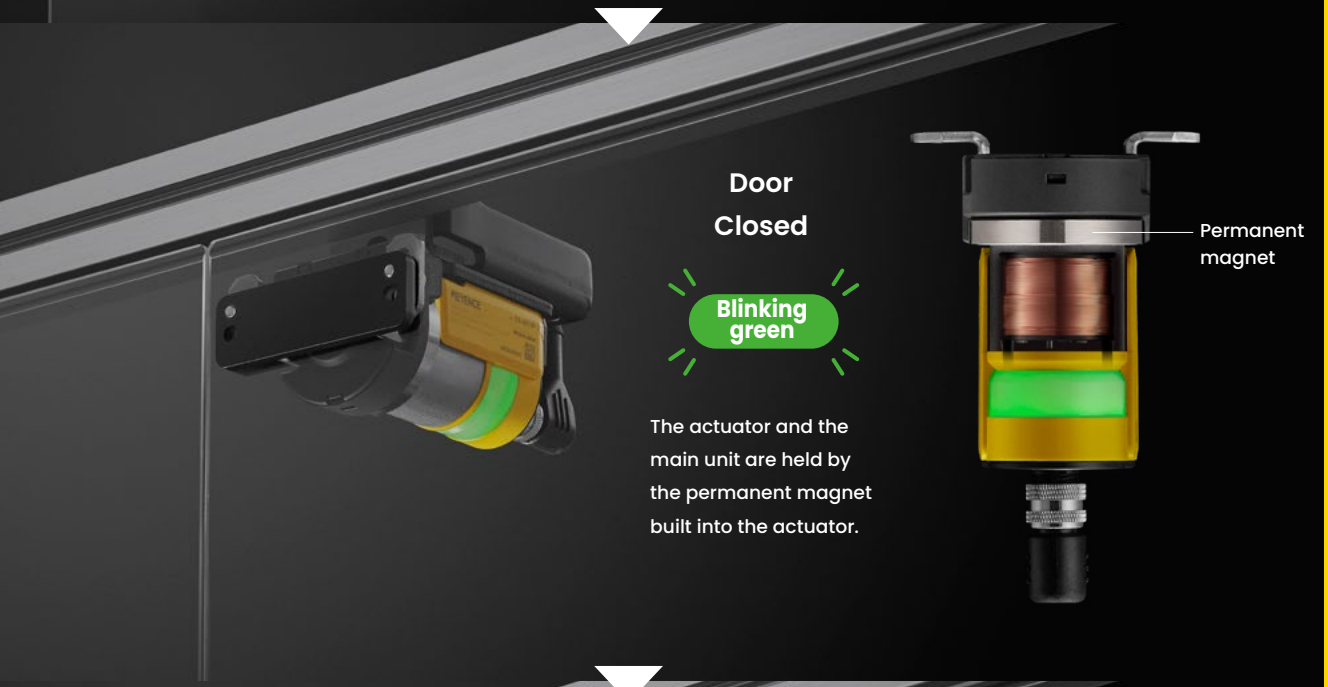
NEW Safety Interlock Switches GS-M Series

Easy, Reliable Locking with Electromagnetic Force



Door Open

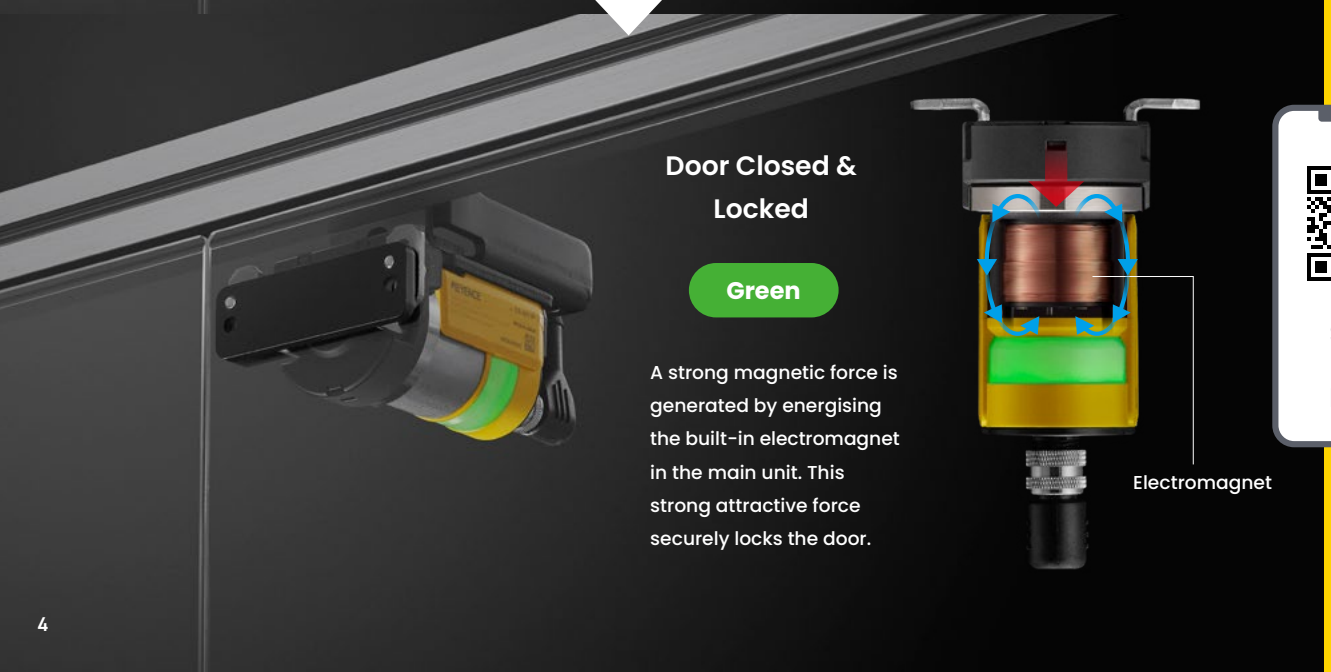
Red



Door Closed

Blinking green

The actuator and the main unit are held by the permanent magnet built into the actuator.



Door Closed & Locked

Green

A strong magnetic force is generated by energising the built-in electromagnet in the main unit. This strong attractive force securely locks the door.

Scan for More

Three functions revolutionising interlock switches

01

→ P. 6

Versatile Mounting



The door is locked by the flat electromagnet and the actuator. The mechanism absorbs door misalignment, eliminating the need for fitting design and enabling easy installation. Ideal for long-term use.

02

→ P. 8

Intuitive Design



Can be installed anywhere without intruding on operation space. An ultra-compact, easy-to-use interlock switch that anyone can use.

03

→ P. 10

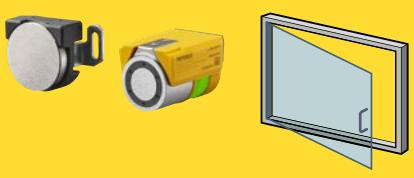
Highly Visible Indicators



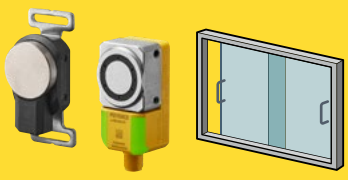
The open, close or error status of the door is indicated by highly visible indicators that can be identified at a glance from anywhere, even from afar.

Select from two types to match the door

GS-M5 (500 N)/**GS-M9** (900 N) **for hinged doors**



GS-ML5 for sliding doors (500 N)



Versatile Mounting



Locks Firmly by Electromagnetic Force

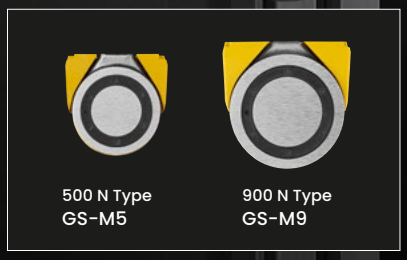
The GS-M Series has a simple, key-less structure, and securely locks using the power of electromagnetic force. It is also highly resistant to door misalignment.



Flush Design

No risk of foreign particles getting stuck

The GS-M Series has a flush design on both the sensor and actuator that allows for reliability in mating and prevents build up or objects from sticking between the devices. If build up does occur, cleaning the device is as simple as a wipe of a rag.



Strong locking force of 500 N/900 N

Holds firmly even without a Actuator key

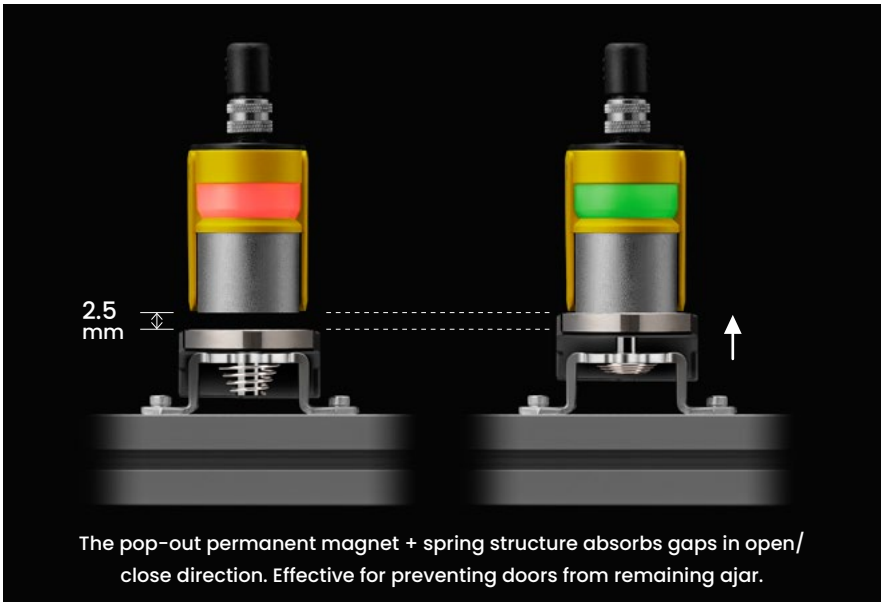
Two models are available for hinged doors in the GS-M Series: 500N and 900N holding force. This allows for flexibility in interlock selection regardless of door size, weight, or application needs in order to keep the door shut and locked.

* Sliding door model features 500N holding force



Flexible, misalignment-resistant actuator

Scan for More



The pop-out permanent magnet + spring structure absorbs gaps in open/ close direction. Effective for preventing doors from remaining ajar.



Supports a tilt of ±4°

Can be installed without worrying about the angle

A spring built into the actuator absorbs door tilt. Easy installation and long reliable use without concerning about installation accuracy.



Supports lateral misalignment of ±5 mm

Door can be locked even when misaligned

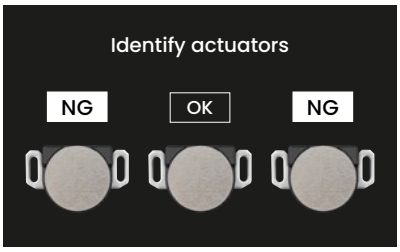
The actuator is designed to be 5 mm larger in the GS-M Series. This design allows the door to be locked securely even if it is misaligned.



Built-in permanent magnet

No magnetic catches required

A permanent magnet is built into the actuator. The door can be held with a constant magnetic force, even without using a magnet catch when it is not locked.



Sensor and Actuator Pairing

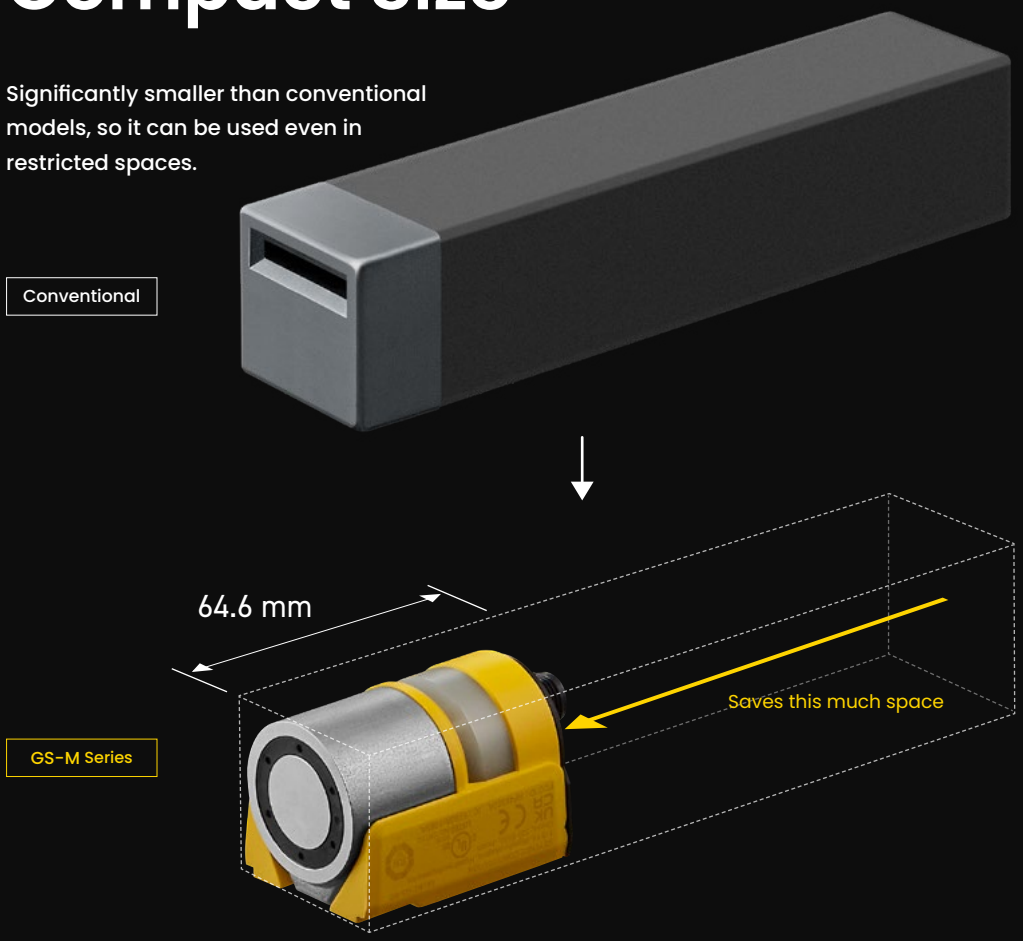
Available two levels of actuator coding

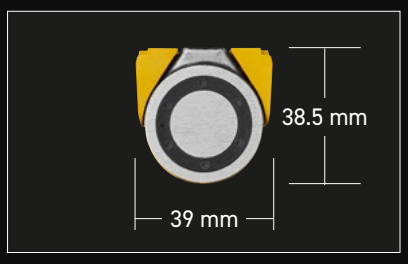
On LOW, the unit will recognize any GS-M actuator that it encounters. On HIGH, the unit can be set to only recognize one specific actuator to prevent tampering, confusion, or operator bypass.

Intuitive Design

An extremely Compact Size

Significantly smaller than conventional models, so it can be used even in restricted spaces.





Smallest in its class: 39 mm

Can be installed anywhere

The detection surface of 39 mm is the smallest in its class. Installation location is no longer a concern in equipment that requires miniaturisation.

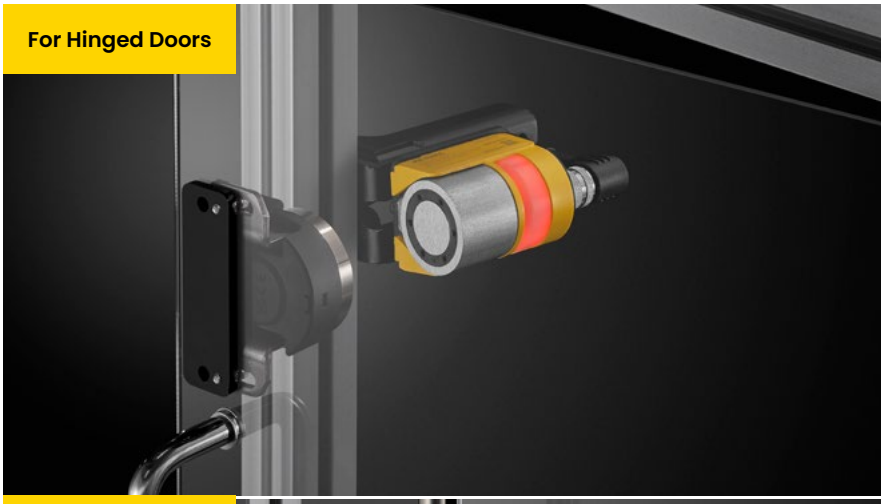


Small footprint

Not obstructing the opening

Significantly smaller than conventional models so that it can be used in areas of the machine is limited.

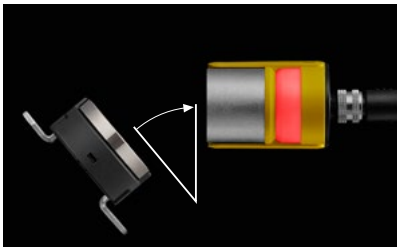
Can be used on a variety of doors without getting in the way





Available in two forms to match the door

There are different models to accommodate for hinged doors or sliding doors. Both models are equipped with the same features that set this hybrid interlock apart from conventional models. The only difference is the physical footprint and how you mount the sensor.



Minimum door turning radius of 45 mm

Suitable for small doors

Small doors such as observation windows that could not be supported by operation key type interlock switches are also suitable for use.



L-shaped cable

No protrusion inside the device

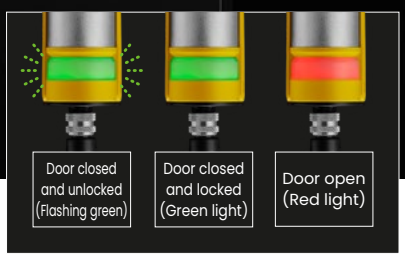
L-shaped and straight cables are available. Cables do not protrude behind the main unit, saving on space including cable space.

Highly Visible Indicators

Cleary recogniza the open/close status

Easy and flexible integration through Wiring Saving units and mixed cascading of GS-M/GS Series.





Lighting patterns according to status

Early detection of abnormalities

The status of the door, including open, closed, locked and in error (flashing red), is indicated. Open doors can be spotted at a glance, shortening equipment downtime.

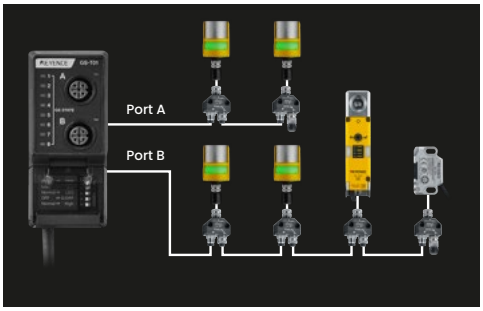


Multi-angle

High visibility from anywhere

The large indicator can be easily seen from any angle. Whether looking up from below or looking from the side, the status of the door can be immediately identified.

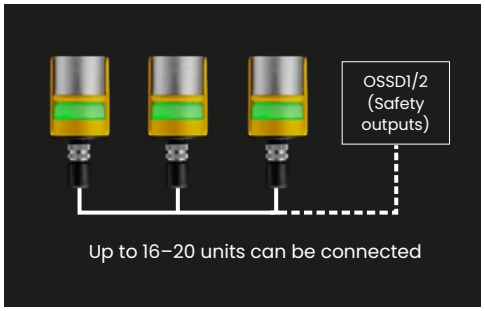
Additional Features



Wiring Saving Unit

Coupling wiring with simplified wiring unit supported

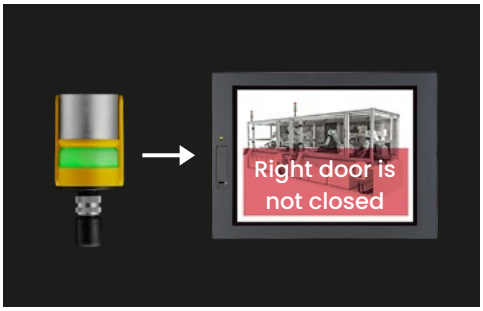
The Wiring Saving Unit reduces installation time and headaches when using multiple interlocks on a machine. Up to 8 GS-M/GS Series interlocks can be cascaded together using Y-shaped connectors and still allow for 8 AUX outputs for door status feedback.



Cascade Multiple Units

Unifying safety outputs

Connect together safety interlocks throughout the machine and cut the total number of safety outputs down to a single pair of OSSDs. It is possible to cascade together GS-M Series and GS Series door interlocks.



Built-In Aux Outputs

Status can be identified remotely

Even when several units are cascaded together, it is possible to quickly identify which access point is open using an additional auxiliary output from each safety interlock. These outputs can be tied to a PLC, HMI, light, etc. to easily identify the open access point.



IP65/67

Can be used in any environment

Achieves an enclosure rating of IP65/67. Suitable for various environments with its best-in-class dust- and water-proof structure. Cleaning the surface is also easy due to the flat structure.



Terminal block type

Any cable can be used

A type with a terminal block at the end is also available. Choose any length or cost of cable that you are already using. If a non-contact type is used, the wiring layout can be used as is.

Selecting a Safety Interlock Switch

STEP 1 Select the appropriate model and actuator

	Hinged door			Sliding door
	Compact [terminal block]	Standard model	High holding force model	
Main Unit				
Actuator				

STEP 2 Select M12 connector cables

M12 L-shaped to M12



M12 to M12



M12 loose wire



M12 8-pin to 5-pin connector (Optional)



STEP 3 Select appropriate brackets

Use the GS-M with the dedicated mounting bracket attached to the main unit. Otherwise, ensure the main unit is installed on a metal surface that allows for heat dissipation (e.g. aluminum). If the power is turned on without such measures and left in a locked state for a long time, the metal parts will become hot.

Standard model



High holding force



Sliding door model



STEP 4 Select Wiring Saving Unit (optional)

Wiring Saving Unit



Y-shaped connector



End terminal



STEP 1 Select the appropriate model

Main Unit

Locking type Power-to-lock type

			Standard model GS-M5 Series			High Holding force model GS-M9 Series		Sliding door model GS-ML5 Series	
									
			Terminal block	Standard type	Advanced function type	Standard type	Advanced function type	Standard type	Advanced function type
M12 Connector type	OSSDs	PNP	GS-M50P	GS-M51P	GS-M53P	GS-M91P	GS-M93P	GS-ML51P	GS-ML53P
		NPN	GS-M50N	GS-M51N	—	GS-M91N	—	GS-ML51N	—
Switching OSSDs	Lock link mode		✔	✔	✔	✔	✔	✔	✔
	Open/Close link mode		✔	✔	✔	✔	✔	✔	✔
AUX	Lock link mode		—	—	✔	—	✔	—	✔
	Open/Close link mode		✔ *1	✔ *1	✔	✔ *1	✔	✔ *1	✔
Cascade	Connect multiple units in series		✔	✔	✔	✔	✔	✔	✔
Y-shaped connector	Easy to use connector for cascading		—	—	—	—	—	—	—
Wiring saving unit	Reduce wiring man hour		—	✔	—	✔	—	✔	—
Interlock	Enable manual reset		—	—	✔	—	✔	—	✔
EDM	Monitor external devices for faults		—	—	✔	—	✔	—	✔
Connection method	M12 connector		—	8-pin	12-pin	8-pin	12-pin	8-pin	12-pin
	Terminal block*2		✔	—	—	—	—	—	—
Switching cording level	Control actuator pairing		✔	✔	✔	✔	✔	✔	✔
Holding force	Locked		500N			900N		500N	
	Unlocked		approx. 30N			approx. 50N		approx. 30N	
Lock control input	Number of locking signals required		1	1	1	1	1	1	1

Actuator

									
Coding level	Low		GS-MA5M			GS-MA9M		GS-MAL5M	
	High		GS-MA5U			GS-MA9U		GS-MAL5U	

*1 When set the OSSD operation to "Open/close link", the AUX cannot be used. *2 Please provide cables and cable glands separately.

Point (1) The GS-M Series locking function is designed for process protection by unlocking when the power is off.
If the door must remain locked even when the power is off, use the GS Series spring lock type. → Page 24

Point (2) Use the GS Series non-contact type if the locking function is not required. → Page 24

STEP 2 Select the M12 connector cable

Total length must be 31 m or less.
Not required for the terminal block type

Standard type (8-pin)

Appearance	Length	Model	Weight
L-shaped (female) - loose wire 	3 m	**GS-P8L3**	Approx. 160 g
Straight (female) - loose wire 	3 m	**GS-P8C3**	Approx. 160 g
	5 m	**GS-P8C5**	Approx. 230 g
	10 m	**GS-P8C10**	Approx. 420 g
L-shaped (female) - straight (male) 	1 m	**GS-P8LC1**	Approx. 80 g
Straight (male) - straight (female) 	1 m	**GS-P8CC1**	Approx. 70 g
	3 m	**GS-P8CC3**	Approx. 150 g
	5 m	**GS-P8CC5**	Approx. 240 g
	10 m	**GS-P8CC10**	Approx. 450 g

Advanced function type (12-pin)

Appearance	Length	Model	Weight
L-shaped (female) - loose wire 	3 m	**GS-P12L3**	Approx. 170 g
Straight (female) - loose wire 	3 m	**GS-P12C3**	Approx. 170 g
	5 m	**GS-P12C5**	Approx. 250 g
	10 m	**GS-P12C10**	Approx. 480 g
	20 m	**GS-P12C20**	Approx. 950 g
L-shaped (male) - straight (female) 	1 m	**GS-P12LC1**	Approx. 85 g

Connector for conversion from standard type (8-pin) to 5-pin

Only available for GS-M51P, GS-M91P and GS-ML51P

Appearance	Length	Model	Weight
Straight (male) Straight (female) 	—	**GS-CC1**	Approx. 19 g

STEP 3 Select mounting brackets

GS-M5/GS-M9

Hinged door

If the frame to which the main unit and the actuator are installed are the same size

When the door is open



When the door is closed



For GS-M5
Frame size
30 to 40 mm **GS-MB11**
40 to 45 mm **GS-MB13**



For GS-M9
Frame size
30 to 40 mm **GS-MB21**
40 to 45 mm **GS-MB23**



GS-M5/GS-M9

Hinged door

- If the door to which the actuator is installed does not have a frame
- If the frame to which the actuator is installed is smaller than the frame to which the main unit is installed

When the door is open



When the door is closed



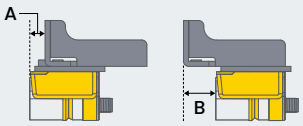
For GS-M5
Frame size on the main unit side
30 to 40 mm **GS-MB12**



For GS-M9
Frame size on the main unit side
30 to 40 mm **GS-MB22**



Adjustable installation position of main sensor unit



Model	A	B
GS-MB12	12 mm	28 mm
GS-MB22	10 mm	30 mm

Adjustable at 5 mm pitch from A to B.

GS-ML5

Sliding door

- If the frame to which the main unit and the actuator are installed are the same size
- If the door to which the actuator is installed does not have a frame
- If the frame to which the actuator is installed is smaller than the frame to which the main unit is installed

When the door is open



When the door is closed



Frame size on the main unit side
30 to 40 mm **GS-MB31**

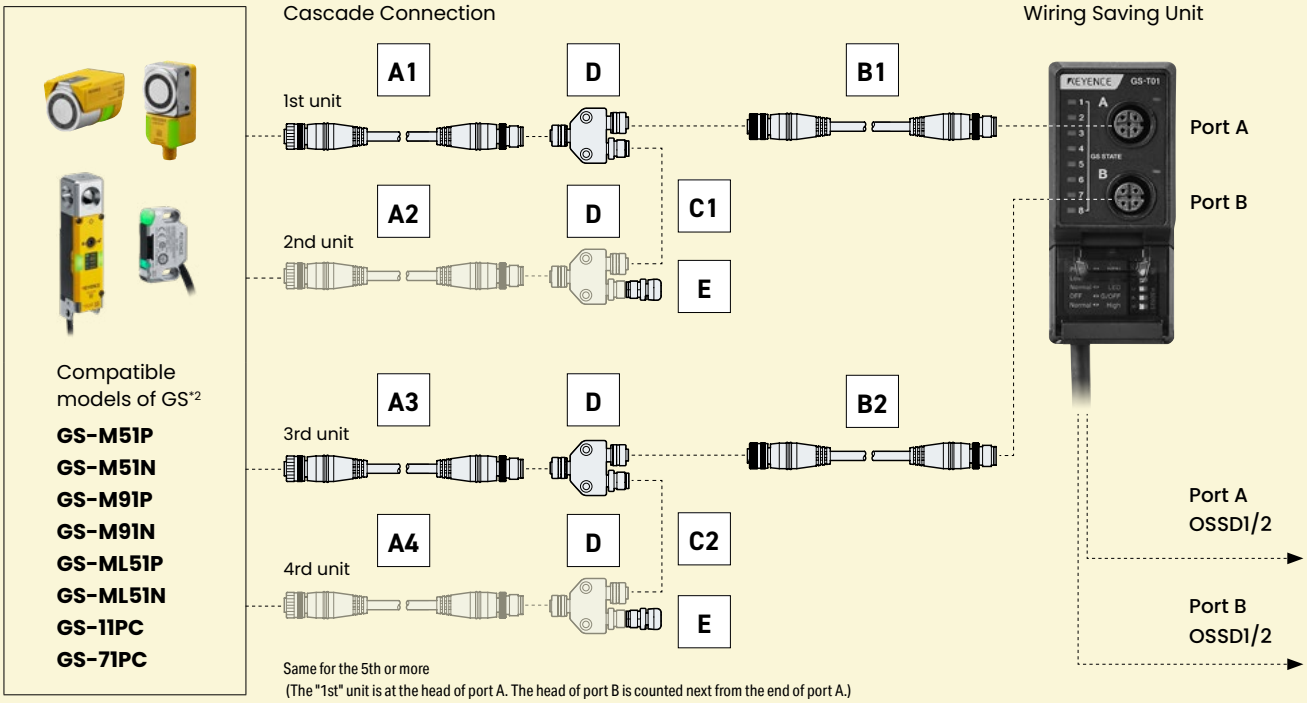


STEP 4 Select Wiring Saving unit (optional)

Wiring Saving Unit GS-T01 (M12 connector, standard type only)

- Equipped with 2 ports and each port has 2 OSSDs. (PLe, Category 4 compliant)
- Up to 8 units of GS-M/GS Series can be cascaded per port, and up to 8 units of them can be cascaded when using 2 ports.
- ON/OFF status of 8 units of GS-M/GS Series can be output individually. (AUX output 1-8 *1)
- Port A and Port B can be locked and controlled individually.

Configuration



*1 AUX output is not for safety. *2 OSSDs of the GS-M/GS Series should be unified as PNP or NPN.

A1 to A8: Connection Cable [M12, 8-pin]

	Length	Model	Weight
	1 m	**GS-P8LC1**	Approx.80 g
	1 m	**GS-P8CC1**	Approx.70 g
	3 m	**GS-P8CC3**	Approx.170 g
	5 m	**GS-P8CC5**	Approx.240 g

D:Y-Shaped Connector, E:End Terminal for Y-Shaped Connector

	Description	Model	Weight
	Y-Shaped Connector	**GS-Y11**	Approx.50 g
	End Terminal for PNP models	**GS-Y12**	Approx.10 g
	End Terminal for NPN models	**GS-Y13N**	Approx.10 g

B1, B2, C1 to C7:Connection Cable [M12, 5-pin]

	Length	Model	Weight
	1 m	**GS-P5CC1**	Approx.90 g
	3 m	**GS-P5CC3**	Approx.210 g
	5 m	**GS-P5CC5**	Approx.310 g
	10 m	**GS-P5CC10**	Approx.580 g

Limitation of Cable Length

Combination of cables		Maximum cable length
A(A1 to A8)		5 m
The port to 1st GS-M/GS	Example: "A1 + B1", "A3 + B2"	31 m
The GS-M/GS to the next GS-M/GS	Example: "A1 + C1 +A2", "A3 + C2 + A4"	31 m

When connecting 3 or more units of the GS-M/GS to a single port, the maximum cable length from the port to the farthest the GS-M/GS is limited as follows.

3 units	4 units	5 units	6 units	7 units	8 units
75 m	55 m	45 m	40 m	35 m	30 m

GS-M Series

OSSD	PNP	**GS-M50P**	**GS-M51P**	**GS-M53P**	**GS-M91P**	**GS-M93P**	**GS-ML51P**	**GS-ML53P**
	NPN	**GS-M50N**	**GS-M51N**	–	**GS-M91N**	–	**GS-ML51N**	–
Type		For hinged door					For sliding door	
		Small			High-holding force			
		Terminal block	M12 Connector		M12 Connector		M12 Connector	
			Standard	Advanced function	Standard	Advanced function	Standard	Advanced function
Holding force when locked		500N			900N		500N	
Holding force when unlocked *1		Approx. 30N			Approx. 50N		Approx. 30N	
Operating distance *2	Sao(OFF → ON)	0.1 mm						
	Sar(ON → OFF)	15 mm						
Response time (ms) *3	Lock → Unlock	250 ms						
	Unlock → Lock	250 ms						
	Detect → Not detect	20 ms + 2 ms × (number of cascaded unit -1)						
	Not detect → Detect	300 ms + 25 ms × (number of cascaded unit -1)						
Cascading		Max. 20 units			Max. 16 units		Max. 20 units	
Control output (OSSD output)	Output	Transistor outputs × 2						
	Max. load current	150 mA						
	Residual voltage (during ON)	Max. 2.5 V (with a cable length of 5 m) Max. 3.5 V (with a cable length of 31 m)						
	OFF state voltage	Max. 2.0 V (with a cable length of 5 m) Max. 3.0 V (with a cable length of 31 m)						
	Leakage current	Max. 0.5 mA						
	Max. capacitive load	0.8 uF						
	Load wiring resistance	Max. 2.5Ω						
AUX (Non-safety-related output)	Output	Transistor output (Only advanced function type has two outputs.)						
	Max. load current	50 mA						
	Residual voltage (during ON)	Max. 2.5 V(with a cable of 5 m) Max. 3.5 V(with a cable of 31 m)						
External input (Short circuit current)	Safety input	Approx. 1.5 mA × 2						
	Lock control input	Approx. 2.5 mA						
	Reset/EDM input	Approx. 5 mA (For advanced function type only)						
	OSSD operation switchin input	Approx. 2.5 mA (For advanced function type only)						
Power supply	Power voltage	24 V DC ±20 % (Ripple P-P 10% or less, Class2)						
	Power consumption	5 W			6 W		5 W	
Protection circuit		Reverse current protection, short-circuit protection and surge protection for each output						
Environ-mental resistance	Enclosure rating	IP65/67 (IEC60529)						
	Operating ambient temperature	-20 to 55℃(No freezing)						
	Storage temperature	-25 to 70℃(No freezing) *4						
	Operating relative humidity	5% to 95%RH						
	Storage relative humidity	5% to 95%RH						
	Vibration resistance	10 to 55 Hz, Double amplitude 2.0 mm, 5 minutes in each of the X, Y, and Z directions (IEC 60947-5-3)						
	Shock resistance	30 G in X, Y, Z directions 6 times each axis (IEC 60947-5-3)						
Applicable Standards (Safety)		EN61508, IEC61508(SIL3), EN ISO13849-1:2015(PL e, Category 4), EN ISO14119(Type4), IEC60947-5-3, EN60947-5-3, UL 60947-5-2						
Material	Main unit	PBT, PET/PAR, TPC, PC, Nickel-plated steel						
	Actuator	PBT, SUS304, Nickel-plated steel			PBT, Painted steel, Nickel-plated steel		PBT, SUS304, Nickel-plated steel	
	Mounting bracket	Aluminum (Plate and screw: Steel)					Aluminum, Painted steel (screw: Steel)	
Weight	Main unit	Approx. 270 g	Approx.250 g	Approx.250 g	Approx.480 g	Approx.480 g	Approx.240 g	Approx.240 g
	Actuator	Approx.160 g			Approx.250 g		Approx.130 g	

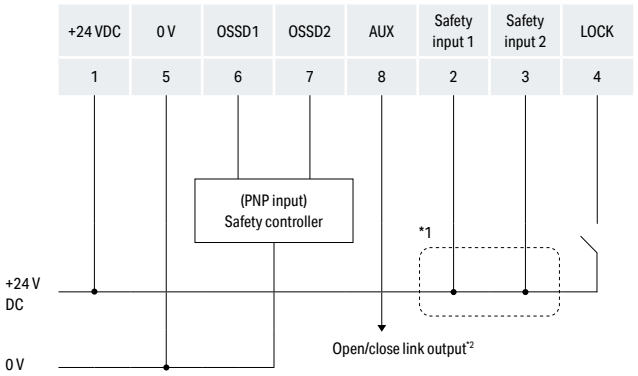
*1 When the actuator is pulled first after unlocked, the holding force increases.
*2 Due to the flexible structure of the actuator, the guard can open 3.8 mm larger than the "Sar". The clearance between the actuator and the main unit may be narrowed by a maximum of 2.3 mm due to the magnet power built into the actuator.
*3 Risk time according to IEC60947-5-3 is 150 ms + 2 ms × (number of cascaded units -1).
*4 When stored for a long period of time, please store it at temperature of 55℃ or lower.

Cables and cable glands for GS-M50P/50N

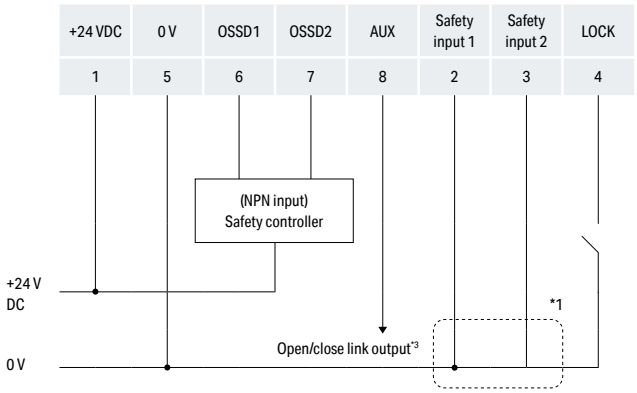
Maximum cable length	31 m
Cable size	AWG 26 to 16
Cable strip length	9 to 10 mm
When ferrule terminals are used	AWG 24 to 18 (maximum outer diameter 1.5 mm) Sleeve length for AWG 24 to 22: 8 to 12 mm Sleeve length for AWG 20 to 18: 12 mm
Cable gland	M16 × 1.5 (male)

Terminal block type

GS-M50P (PNP)



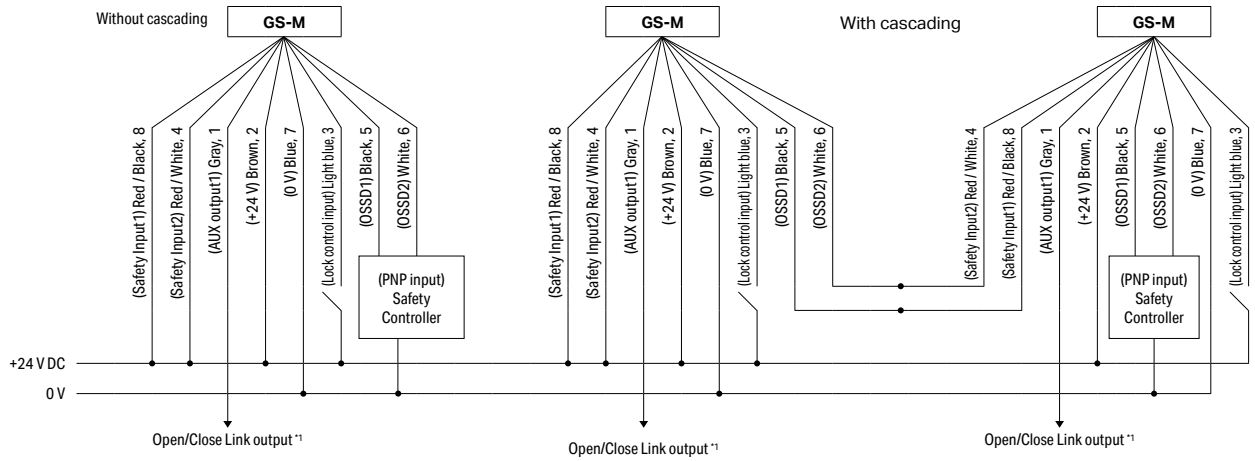
GS-M50N (NPN)



*1 When cascading, safety input 1/2 are connected to the second OSSD1/2. *2 When setting the OSSD operation to open/close link, the AUX output should be connected to +24 V DC.
*3 When setting the OSSD operation to open/close link, the AUX output should be connected to 0 V.

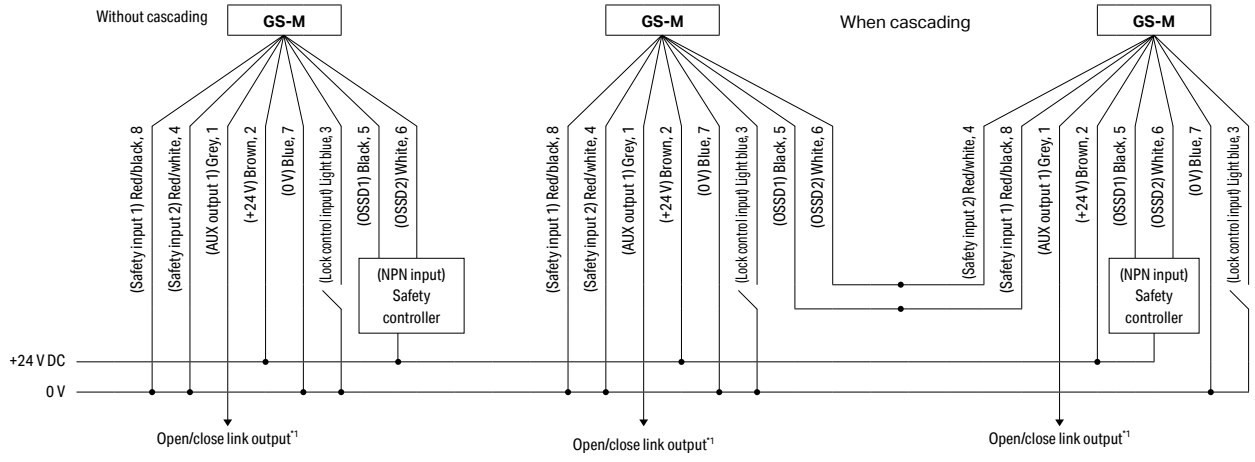
Standard Type

GS-M51P/91P/L51P (PNP)



*1 When set the OSSD operation to “Open/close link”, the AUX should be connected to +24 V DC.

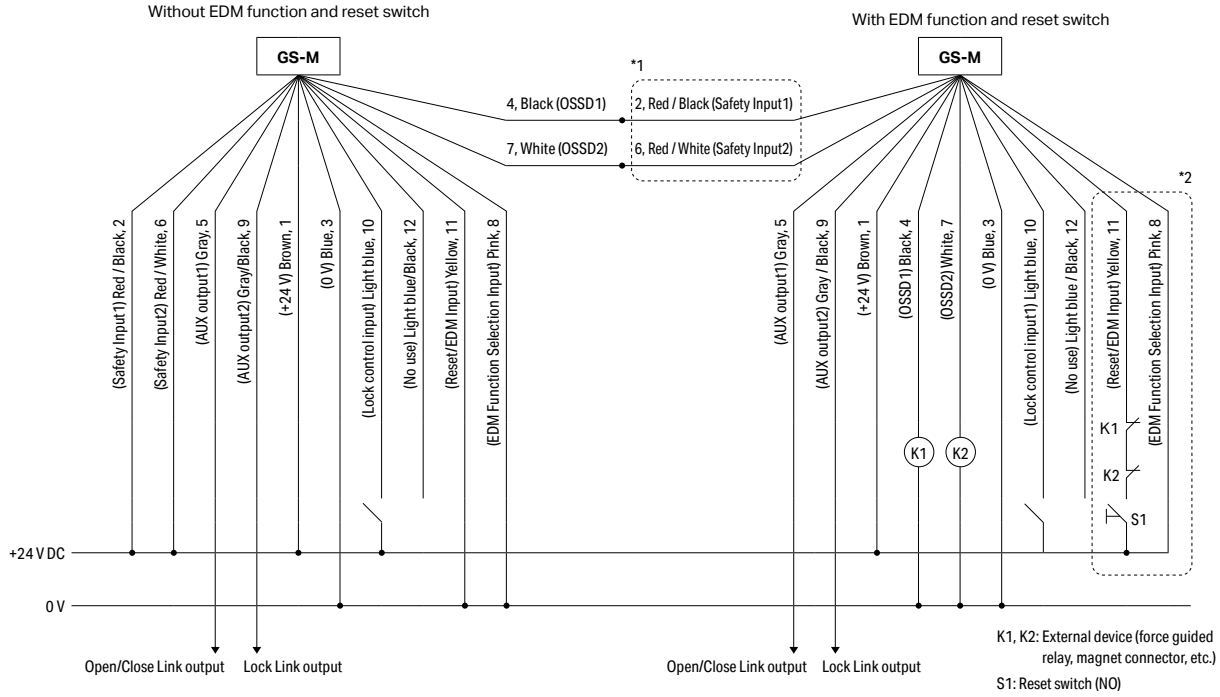
GS-M51N/91N/L51N (NPN)



*1 When set the OSSD operation to “Open/close link”, the AUX should be connected to 0 V.

Advanced function type

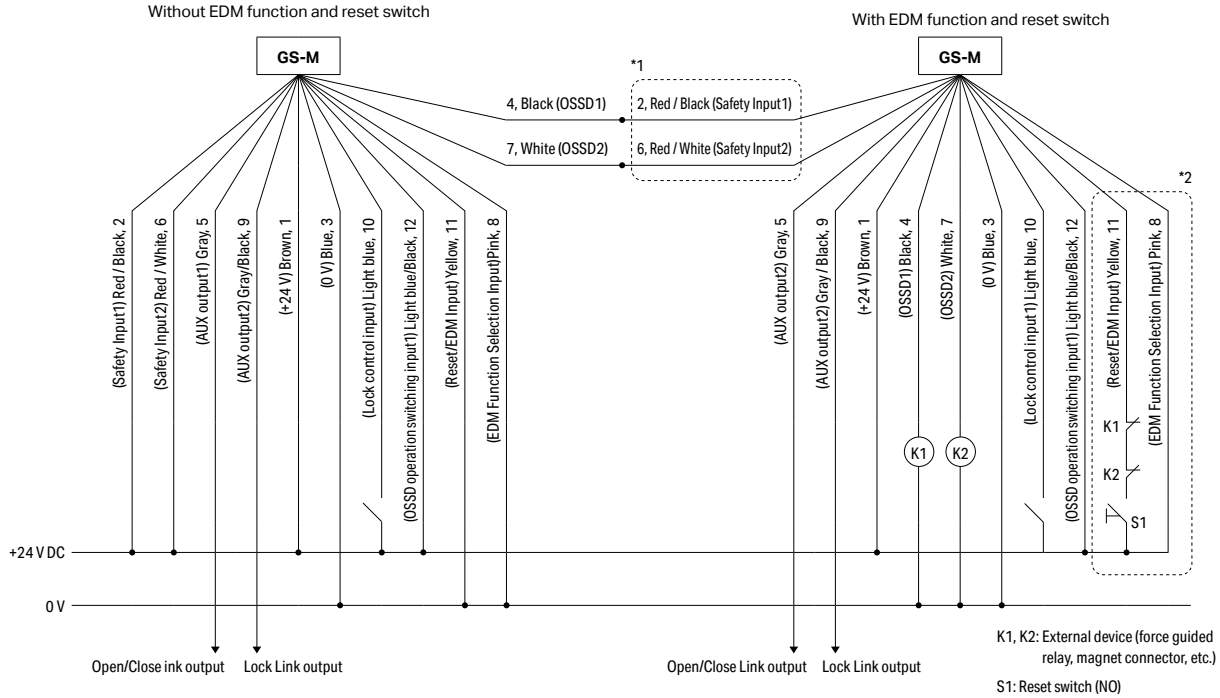
GS-M53P/93P/L53P (PNP) (OSSD operation: Lock link)



*1 When using only 1st unit, the safety input1 and 2 should be connected to +24 VDC.

*2 When using only EDM function, the EDM function selection input should be open and insulated.
When not using both EDM function and reset switch, the EDM function selection input and Reset/EDM input should be connected to 0 V.

GS-M53P / 93P / L53P (PNP) (OSSD operation: Open/close link)



*1 When using only 1st unit, the safety input1 and 2 should be connected to +24 VDC.

*2 When using only EDM function, the EDM function selection input should be open and insulated.
When not using both EDM function and reset switch, the EDM function selection input and Reset/EDM input should be connected to 0 V.

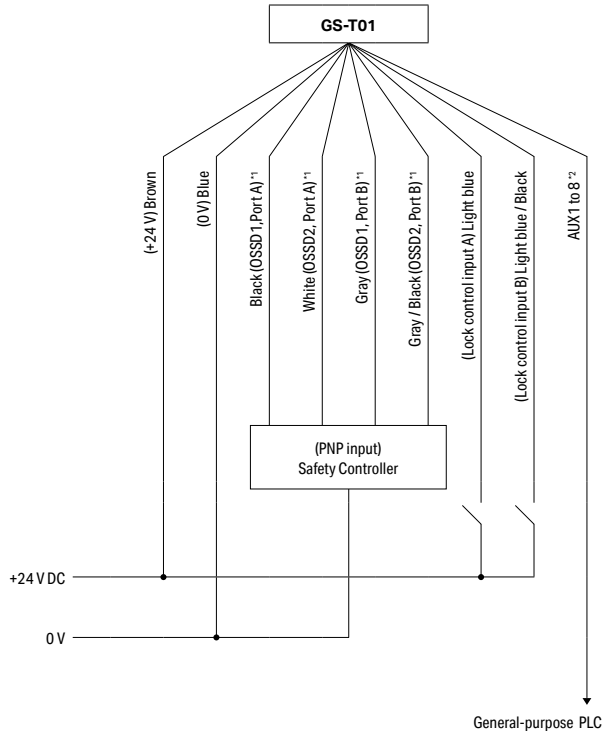
Specifications

Model		GS-T01
Response time (ms)	Lock / Unlock	Response time of GS-M/GS + 110 ms + 70 ms × (number of cascaded unit -1) *1
	OSSD	Response time of GS-M/GS *2
	AUX	Response time of GS-M/GS + 60 ms + 30 ms × (number of cascaded unit -1)
AUX (Non-safety-related output)	Output	Transistor output × 8
	Max. load current	20 mA
	Residual voltage (during ON)	Max. 2.5 V (with a cable length of 2 m)
Lock control input		Approx. 2.5 mA×2
Power supply	Power voltage	24 V DC ±20 % (Ripple P-P 10% or less, Class2)
	Power consumption	0.8 W
Protection circuit		Reverse current protection, short-circuit protection and surge protection
Environmental resistance	Operating ambient temperature	-20 to 55°C(No freezing)
	Storage temperature	-25 to 70°C(No freezing)
	Operating relative humidity	5% to 95%RH
	Storage relative humidity	5% to 95%RH
	Vibration resistance	10 to 55 Hz, Double amplitude 2.0 mm, 5 minutes in each of the X, Y, and Z directions (IEC 60947-5-3)
	Shock resistance	30 G in X, Y, Z directions 6 times each axis (IEC 60947-5-3)
Material		Case: PBT Cable:PVC
Weight		Approx. 290 g

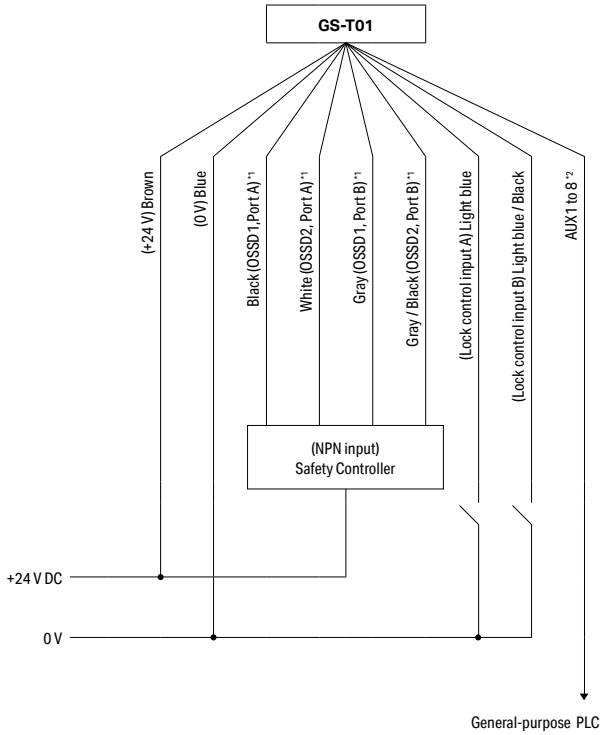
* 1 When using the GS-71PC, the response time from "Unlock" to "Lock" increases by 200 ms per unit.
* 2 When using the GS-71PC, the response time is fixed at 20 ms + 2 ms x (number of cascaded units-1), and the "Sar" is fixed at 10.5 mm.

Wiring Example

For connecting PNP type of GS-M/GS



For connecting NPN type of GS-M

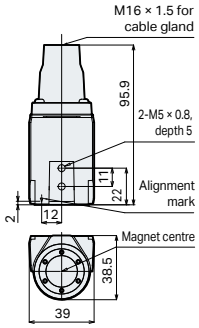


*1 The OSSD1 and OSSD2 of the first GS-M/GS that is connected to the port are output.

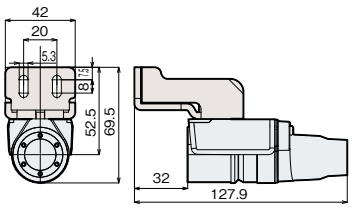
*2 Regardless of the OSSD type (PNP or NPN), the AUX can be used as both NPN (current sinking) output and PNP (current sourcing) output.
When using OSSD and AUX as different types, the operation of AUX will be reversed.

Dimensions Unit: mm

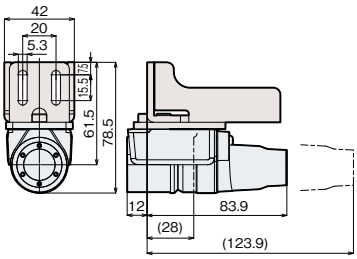
GS-M50



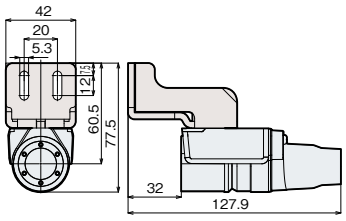
GS-M50 + GS-MB11



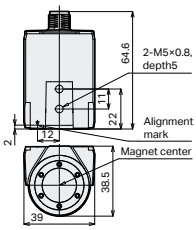
GS-M50 + GS-MB12



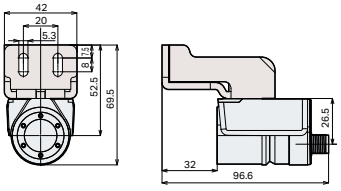
GS-M50 + GS-MB13



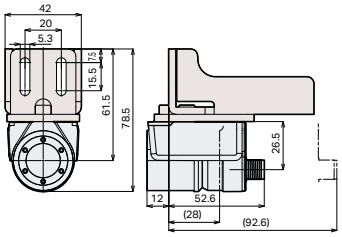
GS-M51/53



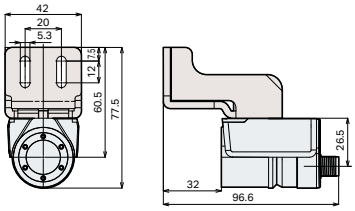
GS-M51/53 + GS-MB11



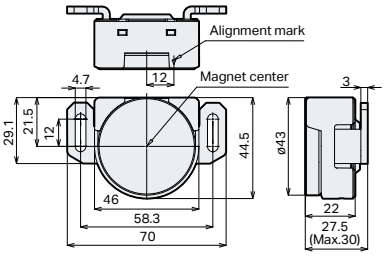
GS-M51/53 + GS-MB12



GS-M51/53 + GS-MB13

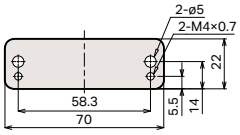


GS-MA5M/U



GS-MB11/12/13

Accessory
(For installing the actuator)



Thickness: 4 mm

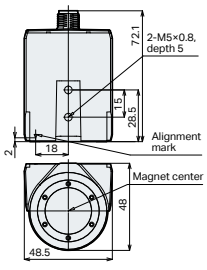
GS-MB11/13 Accessories

M5 hexagon socket head cap screw × 2
across-flats = 4 mm,
L = 8 mm

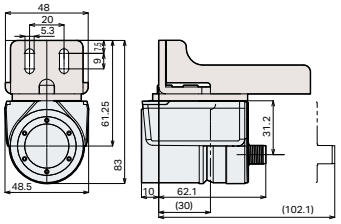
GS-MB12 Accessories

M5 hexagon socket head cap screw × 4
across-flats = 4 mm,
L = 8 mm

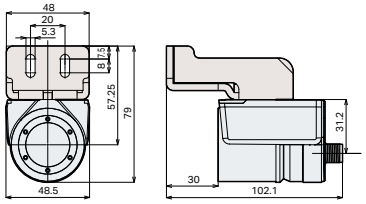
GS-M91/93



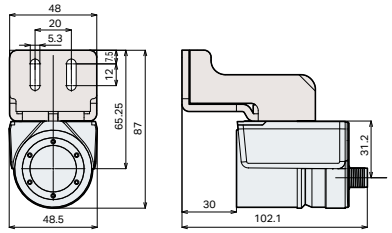
GS-M91/93 + GS-MB22



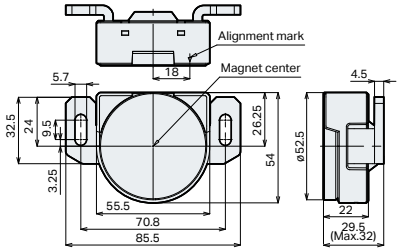
GS-M91/93 + GS-MB21



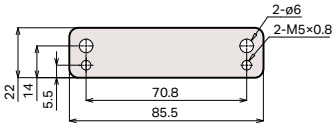
GS-M91/93 + GS-MB23



GS-MA9M/U



GS-MB22 Accessory
(For installing the actuator)



Thickness: 4 mm

GS-MB21/23 Accessories

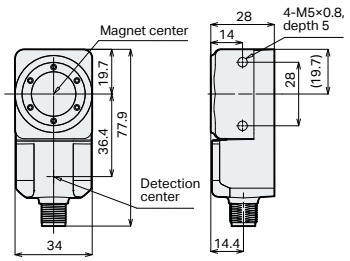
M5 hexagon socket head cap screw × 2
across-flats = 4 mm, L = 8 mm

GS-MB22 Accessories

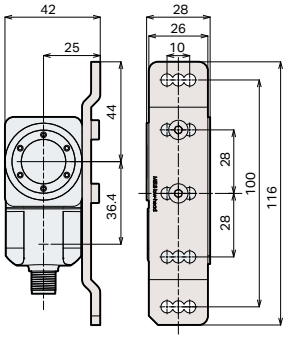
M5 hexagon socket head cap screw × 4
across-flats = 4 mm, L = 8 mm

Dimensions Unit: mm

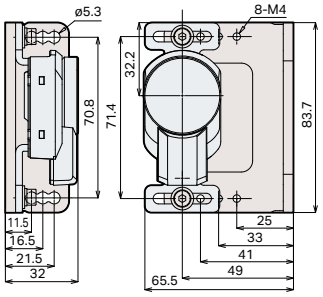
GS-ML51/53



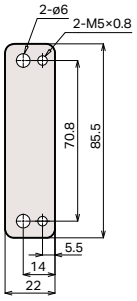
GS-ML51/53 + GS-MB31



GS-MAL5M/U + GS-MB31



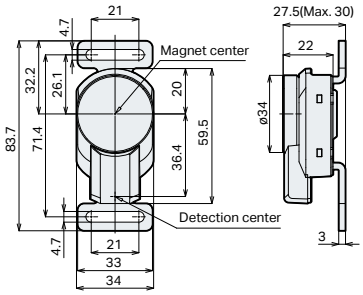
GS-MB31 Accessory
(For installing the actuator)



Thickness: 4 mm

It is possible to install the left and right sides reversed.

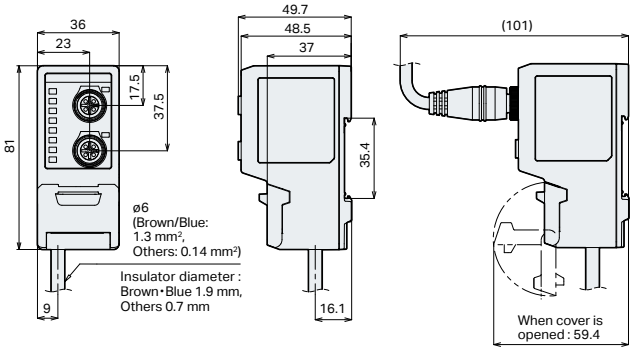
GS-MAL5M/U



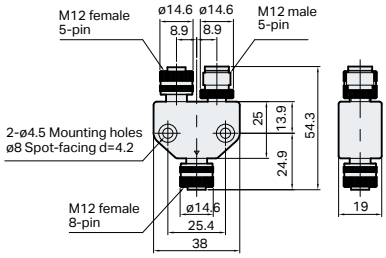
GS-MB31 Accessories

M5 hexagon socket head cap screw × 2, across-flats = 4 mm, L = 8 mm
M4 hexagon socket head cap screw × 2, across-flats = 3 mm, L = 7 mm

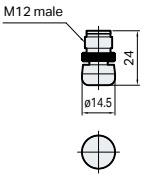
GS-T01



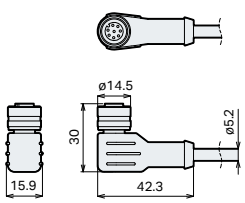
GS-Y11



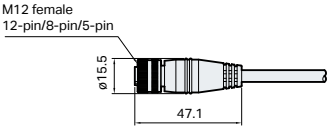
GS-Y12/13N



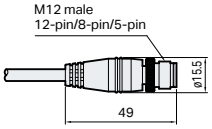
M12 L-shaped connector
(female)



M12 connector
(female)



M12 L-shaped connector
(male)

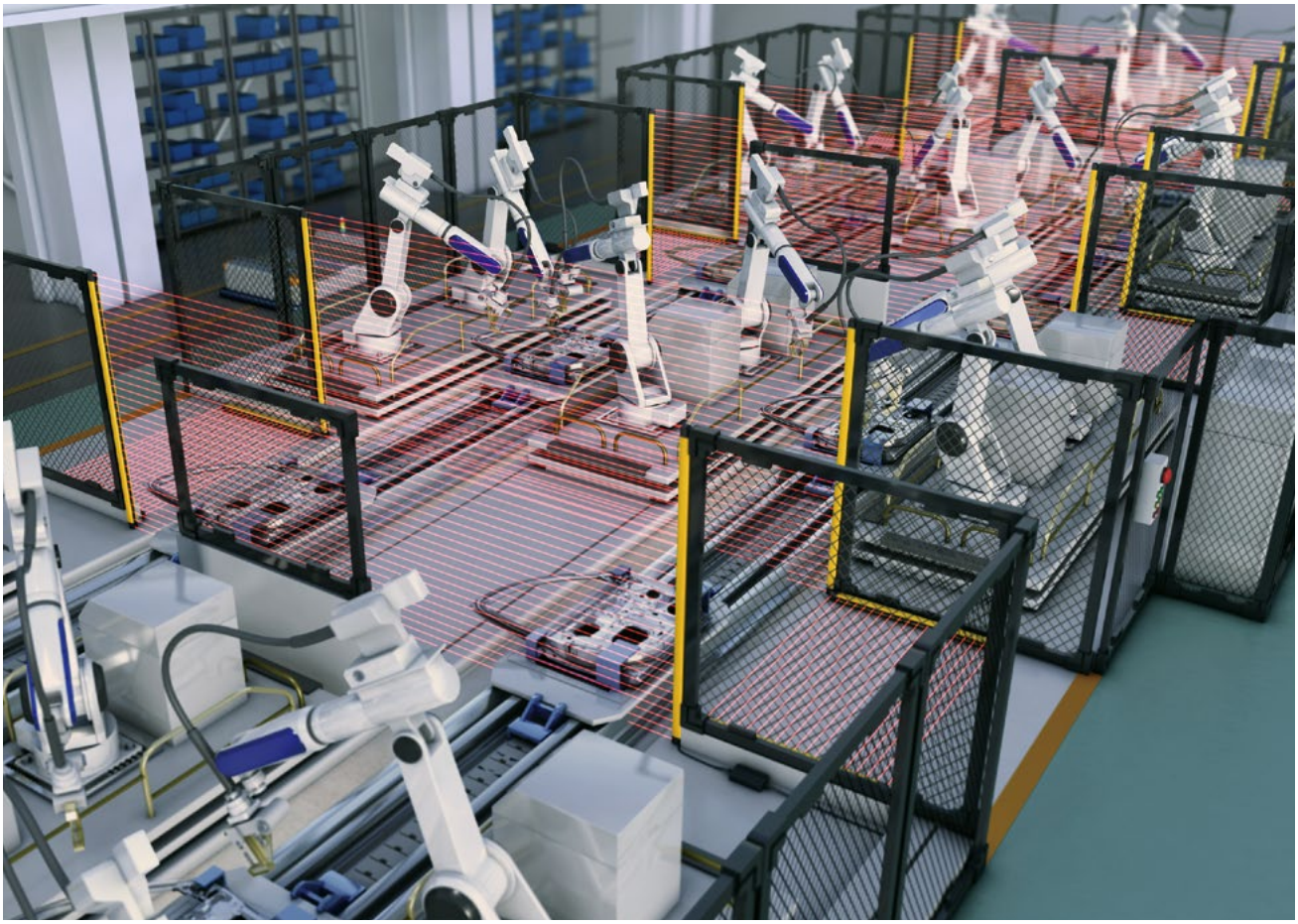


GS-CC1



Cable

Sheath diameter: 5.2 mm Nominal cross-section area: Brown・Blue 0.22 mm², Others 0.14 mm² Insulator diameter: Brown・Blue 1 mm, Others 0.8 mm



Global Standard for Light Curtains



Versatile Lineup

GL-RF Detection capability: ø14 mm
GL-RH Detection capability: ø25 mm
GL-RL Detection capability: ø45 mm

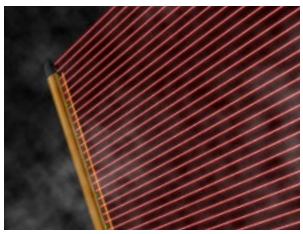
Robust Housing

The GL-R Series features a recessed lens, extruded aluminium housing, and high enclosure ratings to ensure lasting operation in any environment.



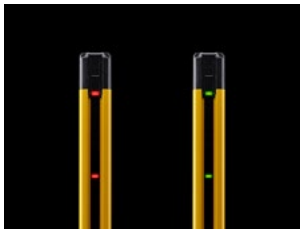
High Powered

Whether detecting over a long range or in an area with high debris and build-up, the GL-R Series provides high power to maintain stable detection and eliminate nuisance trips.



Easy to Align

Alignment has never been easier!
The GL-R Series offers built-in indicators for easy visual alignment, as well as an optional laser alignment tool for more difficult setups.



Locking Type

Prevent Unintended Access to Hazardous Areas



Versatile Lineup

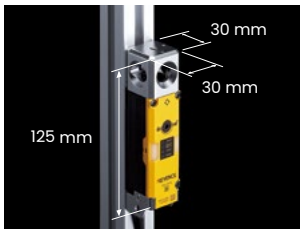
GS-50
Power-to-Release Type

GS-70
Power-to-Lock Type



Compact Size

With a size that is a fraction of conventional models and an impressive 2000N locking strength, these interlocks can be installed virtually anywhere.



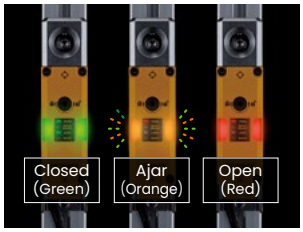
Articulated Actuator

This flexible actuator ensures proper alignment even as doors begin to sag, enabling consistently stable operation on any door.



Highly Visible Indicators

Door status can be recognised with a quick glance. The large indicators can be seen from multiple angles at long distances with ease.



Power-to-Lock



Key Features

- Signals required to engage locking mechanism
- Lock is disengaged if unit power is lost
- Ideal for machines that stop being hazardous once power is lost or removed.

Guard machines that stop immediately upon loss of power.



Power-to-Release



Key Features

- Signals required to release locking mechanism
- Remains locked if unit power is lost
- Ideal for high inertia machines that can remain hazardous after power is lost or removed.

Properly guard machines with long stop times.



Confirm All Access Points are Closed



Versatile Lineup

GS-10
Non-Contact Type



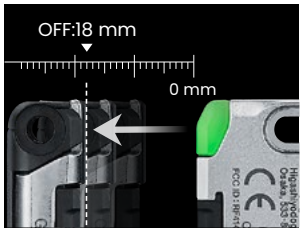
Robust with Small Profile

These models feature a small footprint and metal housing to keep them unobtrusive while also standing up to impacts.



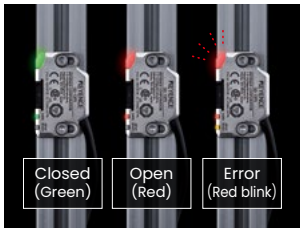
Reliable Detection

Eliminate nuisance trips due to door sag or vibration, as these units offer an impressive detection range to provide consistently stable detection.



Highly Visible Indicators

Confirm access point status from a distance with some of the industry's largest indicators. Easily check the open or closed status, and even see if there is an error.



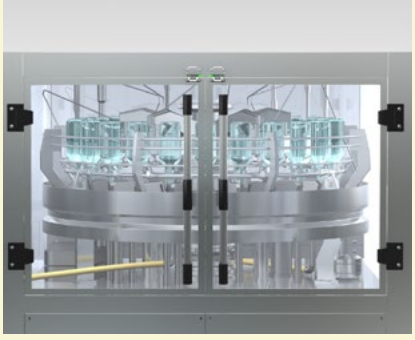
Non-Contact Type



Key Features

- Easily identify door open status
- Contamination resistant design
- Ideal for doors or gates that will remain closed without a locking mechanism
- Also useful for confirming origin position of moving machinery

Door Switch:
Monitor door status without concern for build-up or damage.



Origin Sensor:
Ensure moving components are located in the correct home positions.



Choose the type according to the necessary functions

		Locking type (Power-to-release type) GS-50 Series		Locking type (Power-to-lock type) GS-70 Series		Non-contact type GS-10 Series		
								
		Power-to-release		Power-to-lock		Non-contact type		
		Standard type	High performance type	Standard type	High performance type	Simple function type	Standard type	High performance type
OSSDs	Redundant safety outputs	✔	✔	✔	✔	✔	✔	✔
Switching OSSDs	(Locking Type) OSSDs can be linked to Open/Close status	—	—	—	✔	—	—	—
AUX output(s)	Additional output(s) to monitor status	1	2	1	2*1	1	1	1
Cascade	Connect multiple units in series	✔	✔	✔	✔	—	✔	✔
Y-shaped connector	Easy to use connector for cascading	—	—	—	—	—	✔*2	—
Interlock	Enable manual reset through unit	—	✔	—	✔	—	—	✔
EDM	Monitor external devices for faults	—	✔	—	✔	—	—	✔
Switching encoding level	Control actuator pairing	✔	✔	✔	✔	✔	✔	✔
Lock control input(s)	Number of locking signals required	1	2	1	1	—	—	—
Manual release	Disengage lock through face of unit	✔	✔	✔	✔	—	—	—
Escape release compatible*3	Disengage lock from inside hazardous area	✔*2	✔	✔*2	✔	—	—	—
Handle compatible*4	Integrate with two-action handle	✔	✔	✔	✔	—	—	—

Select a unit based on cable and output preferences

		Power-to-release		Power-to-lock		Non-contact type		
		Standard type	Standard type	Standard type	High performance type	Simple function type	Standard type	High performance type
Standard cable (5 m)	PNP	GS-51P5	—	GS-71P5	—	—	GS-11P5	GS-13P5
	NPN	GS-51N5	—	GS-71N5	—	—	GS-11N5	—
Standard cable (10 m)	PNP	GS-51P10	—	GS-71P10	—	—	GS-11P10	—
	NPN	GS-51N10	—	GS-71N10	—	—	GS-11N10	—
M12 connector type	PNP	GS-51PC	GS-53PC	GS-71PC	GS-73PC	GS-10PC	GS-11PC	GS-13PC
	NPN	—	—	—	—	—	—	—

*1 If you choose open/close link mode for OSSD operation, number of AUX outputs is one. *2 Compatible with M12 connector type only
*3 Compatible with KEYENCE optional escape release (GS-H02) *4 Compatible with KEYENCE optional handle (GS-H01)

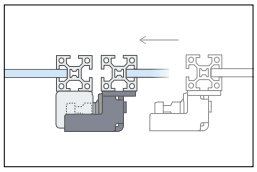
Cables for M12 connector type models

	Type	Length	Model	Weight
Standard 	Simple function type (5-pin)	5 m	GS-P5C5	Approx. 200 g
		10 m	GS-P5C10	Approx. 390 g
	Standard type (8-pin)	5 m	GS-P8C5	Approx. 230 g
		10 m	GS-P8C10	Approx. 420 g
	High performance type (12-pin)	5 m	GS-P12C5	Approx. 250 g
		10 m	GS-P12C10	Approx. 480 g
		20 m	GS-P12C20	Approx. 950 g
For extension 	Simple function type (5-pin)	5 m	GS-P5CC5	Approx. 310 g
		10 m	GS-P5CC10	Approx. 580 g
	Standard type (8-pin)	1 m	GS-P8CC1	Approx. 70 g
		5 m	GS-P8CC5	Approx. 240 g
		10 m	GS-P8CC10	Approx. 450 g

Sliding door

Locking type





For main unit
No bracket

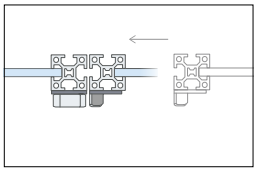
For actuator
Mounting bracket for sliding door (right side slides)
GS-B33*



*Mounting bracket for sliding door (left side slides) GS-B43 is also available.

Non-contact type





For main unit
Flat mounting bracket
GS-B11



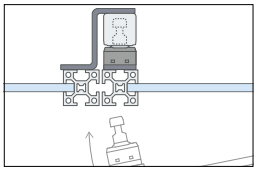
For actuator
Flat mounting bracket
GS-B11



Hinged door (attachment on the inside)

Locking type



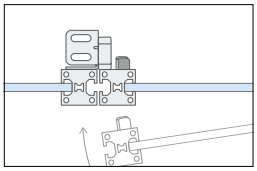


For main unit & actuator
Mounting bracket
(attachment on the inside)
GS-B21



Non-contact type





For main unit
L-shaped mounting bracket
GS-B01

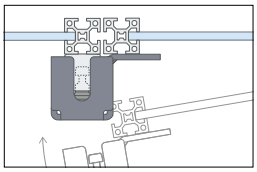


For actuator
No bracket

Hinged door (attachment on the front side)

Locking type





For main unit
No bracket

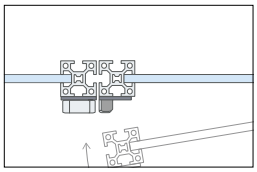
For actuator
Hinged door mounting bracket (right-open)
GS-B31*



*Hinged door mounting bracket (left-open) GS-B41 is also available.

Non-contact type





For main unit
Flat mounting bracket
GS-B11



For actuator
Flat mounting bracket
GS-B11



Locking type accessories (for GS-50/70 series)

Replacement Actuator
GS-A21
Approx. 60 g



Dedicated Handle
GS-H01
Approx. 1070 g



Escape Release
GS-H02
Approx. 90 g



Non-contact type accessories

Replacement Actuator
GS-A01
Approx. 15 g



Y-Shaped Connector
GS-Y01
Approx. 50 g



End Terminal for Y-Shaped Connector
GS-Y02
Approx. 15 g



Other Sensor Line-up

Keyence LR Series can solve any sensor applications

DISTANCE

Medium range

Digital
CMOS Laser Sensor

LR-X
SERIES

Detecting Distance

LR-X50xx
25 to 50 mm

LR-X100xx
25 to 100 mm

LR-X250xx
30 to 250 mm



Self-contained
CMOS Laser Sensor

LR-Z
SERIES

Detecting Distance

LR-ZB100xx
35 to 100 mm

LR-ZB250xx
35 to 250 mm

LR-ZH500xx
35 to 500 mm



Long range

Self-contained
TOF Laser Sensor

LR-T
SERIES

Detecting Distance

LR-TB2000xx
60 to 2000 mm

LR-TB5000xx
60 to 5000 mm

LR-TB5000



LR-TB2000





INTENSITY + COLOUR

Self-Contained
Full-Spectrum Sensor

LR-W
SERIES



Long Range
Model
LR-W500(C)

KEYENCE CORPORATION

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